

Reproducibility: Understanding Impact of Human Feedback via Influence Functions

Anonymous ACL submission

1 Introduction

This project aims to reproduce the main findings of (Min et al., 2025), which studies how influence functions can quantify the impact of individual feedback samples on reward model performance in RLHF. The paper shows that biased preference data, such as preferences favoring verbose or sycophantic responses, can be detected by measuring how much each training sample harms a targeted validation set.

In our reproduction, we will keep the main experimental structure of the paper but adapt the implementation to our available resources. We will use a single RTX 5090 as the primary training device, use API calls to ChatGPT-4o for LLM-based judging and scoring, and prioritize Qwen-based reward models as our main backbone. If time permits, we will additionally evaluate more model backbones for comparison. We also aim to reduce manual operations as much as possible: except for minimal sanity checking if needed, dataset scoring, filtering, and bias annotation will be completed with LLM.

1.1 Hessian Matrix Compression

The first part is the influence computation pipeline. Since exact inverse-Hessian computation is intractable for large reward models, the paper uses DataInf to approximate inverse-Hessian effects and OPORP to compress gradient vectors (Kwon et al., 2024; Li and Li, 2023). We will reproduce this pipeline on our Qwen-based reward model and compare it with a DataInf-only implementation.

The expected result is to recover the main efficiency trend reported in the paper: gradient compression should reduce the storage cost of a single gradient vector from about 160MB to 256KB and significantly speed up influence computation. Our goal is not exact runtime matching, since our hardware differs from the original setup, but rather reproducing the same qualitative gain.

1.2 Dataset Construction

The main dataset for reproduction is Anthropic-HH (Bai et al., 2022). We will reproduce both the length-bias and sycophancy-bias settings from the paper.

For the Length Bias Training set, we will sample 15k instances from Anthropic-HH-helpful and flip labels when the chosen response is shorter than the rejected one, so that the corrupted labels favor verbose responses. For the Length Bias Validation set, we will split the validation data into the Concise subset and the *Verbose* subset following the paper.

For the Sycophancy Bias Training set, we will sample 15k instances from Anthropic-HH-helpful-online and use ChatGPT-4o to score the two responses for sycophancy on a Likert scale. We will then flip labels when the chosen response is less sycophantic than the rejected one. For the Sycophancy Bias Validation set, we will reproduce the *Less Sycophantic* and *More Sycophantic* subsets. Compared with the original paper, we intentionally simplify the pipeline by primarily using one strong API judge instead of relying on multiple human or model annotators, which better matches our goal of reducing manual effort.

1.3 Bias Detection

After training the reward model and computing influence scores, we will compare the influence-based detector with the main baselines used in the paper: Mahalanobis Distance, KNN, Self-confidence, Entropy, and ChatGPT-4o few-shot judging. Evaluation will focus on True Positive Rate, False Positive Rate, and AUC.

Our main target is to reproduce the conclusion of Figure 2: influence functions should outperform standard uncertainty- or distance-based baselines for both length bias and sycophancy bias detection. Because we will use ChatGPT-4o API for scoring and judging, exact numeric agreement with

the original paper is not required; preserving the ranking and the main conclusion is the primary goal.

1.4 Ablation Studies

We will reproduce the main ablation results related to validation-set design. In particular, we will test if targeted validation sets, such as Concise or *Less Sycophantic*, are more effective than mismatched validation sets, as reported in the paper . We will also test whether the method remains effective when the targeted validation set is small.

To keep the project manageable, we will simplify the LLM baseline ablation. Instead of reproducing many variants of multi-model few-shot prompting, we will mainly use ChatGPT-4o as the API baseline and vary only the number of demonstrations when time permits. This keeps the comparison consistent with our limited compute budget and our preference for minimizing manual intervention.

1.5 Detecting Biased Samples in the Original Human Dataset

We plan to reproduce the experiment on the original, uncorrupted human dataset. The paper shows the top-ranked harmful samples identified by influence scores contain substantially more mislabeled or problematic examples than random samples .

Because our project aims to reduce manual work, we will use ChatGPT-4o as the main reviewer for the Top- k suspicious samples and compare them with random samples under the same review protocol. If time allows, we may manually inspect a small subset as a sanity check, but the primary pipeline will remain LLM-driven. Our goal is to verify the same qualitative conclusion: influence-guided filtering is useful for practical data curation.

1.6 Labeling Strategy Oversight

For labeling strategy oversight, we will follow the HelpSteer2 setup in the paper (Wang et al., 2024). The training split will be used to construct Bob’s dataset D_B , and the validation split will be used to construct Alice’s expert dataset D_A . Each response has four dimensions: correctness, coherence, complexity, and verbosity.

We will pre-define Alice’s strategy and initialize Bob with a suboptimal strategy, then update Bob’s weights using the influence-guided procedure in the paper. We will report label accuracy, reward model accuracy, and cosine similarity, following Figure 5 (Min et al., 2025; Lambert et al., 2024).

To reduce manual work, Alice and Bob will both be simulated from scoring rules rather than real annotators. This setting is faithful to the paper while remaining practical for our reproduction.

2 Plan for Extension

We plan to extend to the safety domain. Instead of focusing only on verbosity and sycophancy, we want to test whether influence functions can detect safety bias, namely training samples that systematically favor unsafe or weakly aligned responses.

We prefer to use an existing safety RLHF dataset, especially BeaverTails (Ji et al., 2024), because this avoids additional manual annotation. If necessary, we may also reuse Anthropic-HH and generate safety labels with ChatGPT-4o. In both cases, the goal is to test whether influence functions can identify harmful safety-related samples and whether a small expert-style validation set can guide non-expert labeling in the safety setting.

More broadly, our reproduction will use Qwen as the primary reward model backbone, and, if time permits, we will extend the study to additional model families. This will help evaluate whether the paper’s conclusions are specific to one backbone or remain stable across models.

References

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